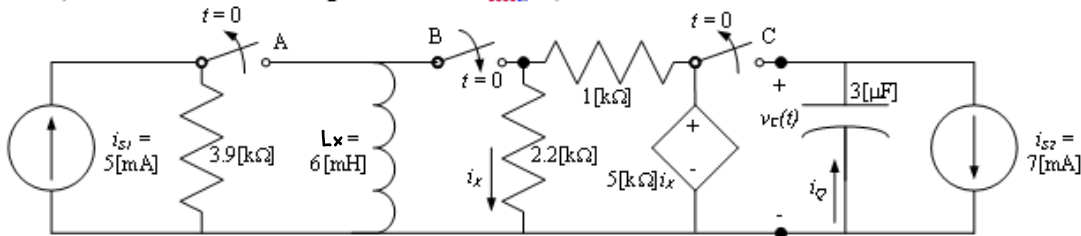


ECE 2300 – Quiz #5

November 10, 2015

Use the circuit below to solve. The switches A and C were closed for a long time and switch B was open for a long time before $t = 0$. Then, at $t = 0$, all three switches changed position instantaneously.

- Find $i_x(0^+)$.
- Find $i_x(10[\mu s])$.
- For 3 extra bonus points, find $i_Q(0^+)$.
- For 3 extra bonus points, find $v_C(0^+)$.



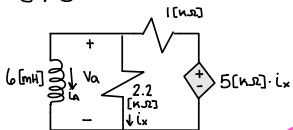
① $t < 0$

Inductors become wires, capacitors open



$$i_A(0) = 5[\text{mA}]$$

② $t > 0$



No independent source; Natural response
a time constant to get $i_A(t)$

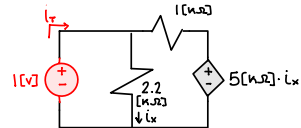
③

Replace inductor with a test source

$$L_x = \frac{1[\text{V}]}{2.2[\text{k}\Omega]}$$

$$i_T = i_x + \frac{1[\text{V}] - 5[\text{k}\Omega]i_x}{1[\text{k}\Omega]}$$

Simultaneously solve
 $i_T = -818[\mu\text{A}]$



$$R_{eq} = \frac{1[\text{V}]}{i_T} = \frac{1[\text{V}]}{-818[\mu\text{A}]} = -1.2[\text{k}\Omega]$$

④ solve for τ

$$\tau = \frac{L_x}{R_{eq}} = \frac{6[\text{mH}]}{-1.2[\text{k}\Omega]} = -4.91[\mu\text{s}]$$

⑤ $t \geq 0$

$$i_A(t) = (5[\text{mA}])e^{-t/4.91[\mu\text{s}]}$$

⑥ $t > 0$

$$v_A = L_x \cdot \frac{di_A}{dt} = \frac{(6[\text{mH}]) (5[\text{mA}])}{4.91[\mu\text{s}]} \cdot e^{-t/4.91[\mu\text{s}]} = 6.11e^{-t/4.91[\mu\text{s}]} [\text{V}]$$

③ $t > 0$

$$i_x(t) = \frac{v_A}{2.2[\text{k}\Omega]} = (2.78[\text{mA}])e^{-t/4.91[\mu\text{s}]}$$

a.

$$i_x(0^+) = 2.78[\text{mA}]$$

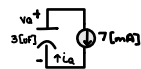
b.

$$i_x(10[\mu\text{s}]) = 2.78[\text{mA}]e^{-10/4.91} = 21.3[\text{mA}]$$

c.

find $i_Q(0^+)$

$t > 0$

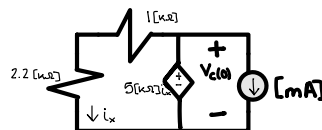


$$i_Q(0^+) = 7[\text{mA}]$$

d.

find $v_C(0^+)$

$t < 0$



$$v_C(0) = 5[\text{k}\Omega] i_x$$

$$i_x = \frac{5[\text{k}\Omega] i_x}{2.2[\text{k}\Omega] + 1[\text{k}\Omega]} = 1.56 i_x \quad ; i_x = 0$$

$$v_C(0) = 0 = v_C(0^+)$$

2. {30 Points} In the circuit below it is given that

$$v_s(t) = -7 \left[\frac{\text{V}}{\text{s}} \right] t,$$

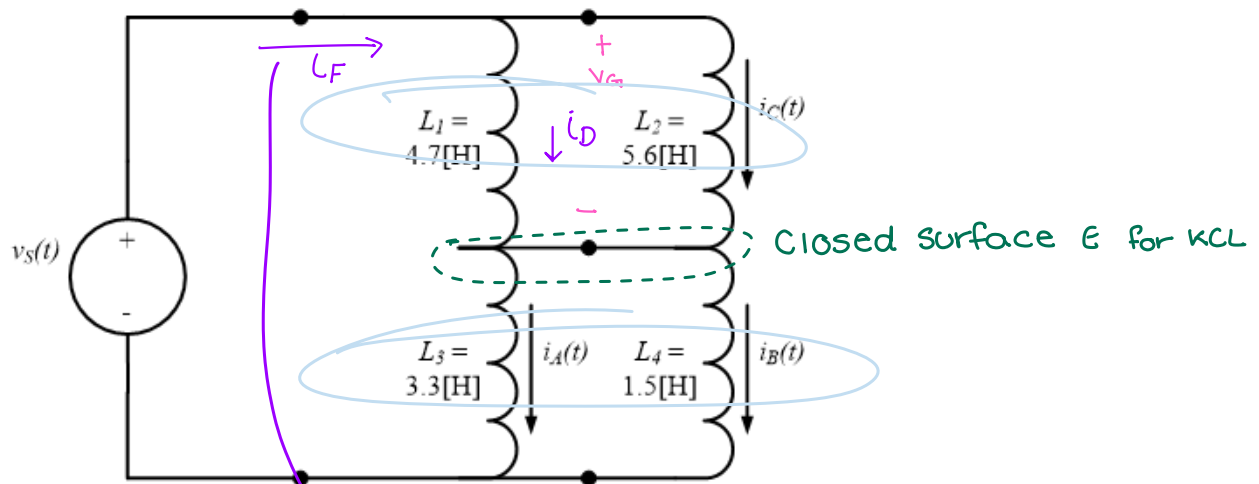
$$i_A(0) = 6.8 [\text{A}],$$

$$i_B(0) = -4.7 [\text{A}], \text{ and}$$

$$i_C(0) = -8.9 [\text{A}].$$

a) Find the energy stored in L_1 at $t = 4[\text{s}]$.

b) Find the power delivered by $v_s(t)$ at $t = 4[\text{s}]$.



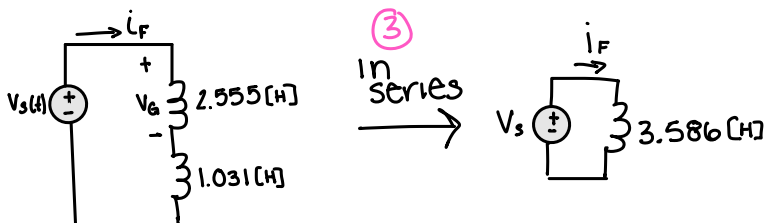
KCL:

$$i_D(0) = i_A(0) + i_B(0) - i_C(0) = 11 [\text{A}]$$

$$L_1 \parallel L_2 = 2.555 [\text{H}]$$

$$L_3 \parallel L_4 = 1.031 [\text{H}]$$

② redraw



$$④ \quad i_F(t) = \frac{1}{3.586 [\text{H}]} \int_0^t v_s(x) dx + i_F(0)$$

$$\rightarrow i_F(0) = i_D(0) + i_C(0) = 2.1 [\text{A}]$$

$$i_F(t) = \left(\frac{-7}{3.586} \right) \cdot \frac{x^2}{2} \Big|_0^t + 2.1 [\text{A}] = -0.976 \left[\frac{\text{A}}{\text{s}} \right] t^2 + 2.1 [\text{A}]$$

$$V_G = 2.555 [\text{H}] \frac{di_F}{dt} = (2.555) (-0.976) (2) t$$

$$V_G(t) = -4.987 \left[\frac{\text{V}}{\text{s}} \right] t$$

Continued

2. {30 Points} In the circuit below it is given that

$$v_s(t) = -7 \left[\frac{\text{V}}{\text{s}} \right] t,$$

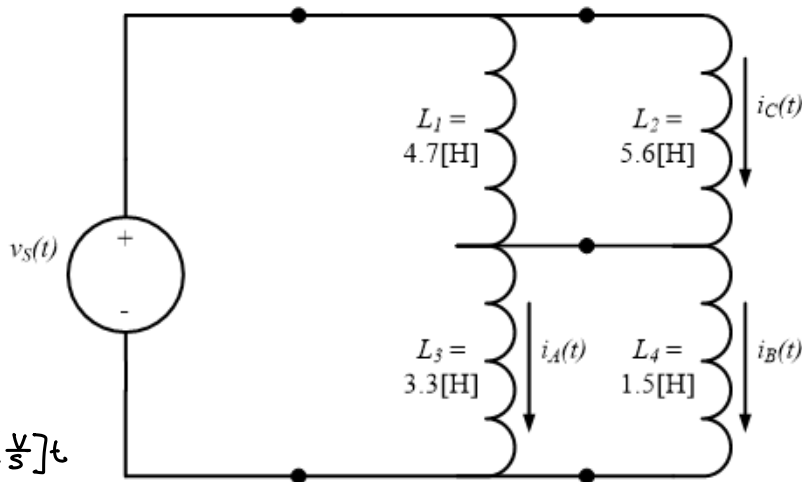
$$i_A(0) = 6.8 [\text{A}],$$

$$i_B(0) = -4.7 [\text{A}], \text{ and}$$

$$i_C(0) = -8.9 [\text{A}].$$

a) Find the energy stored in L_1 at $t = 4[\text{s}]$.

b) Find the power delivered by $v_s(t)$ at $t = 4[\text{s}]$.



$$v_G(t) = -4.987 \left[\frac{\text{V}}{\text{s}} \right] t$$

a.

$$L_D = \frac{1}{4.7[\text{H}]} \int_0^t v_G(x) dx + 11 [\text{A}]$$

$$L_D(4[\text{s}]) = \frac{-4.987}{(4.7)(2)} \cdot (4)^2 + 11 = 2.511 [\text{A}]$$

$$w_{\text{sto}, L_1}(4[\text{s}]) = \frac{1}{2} (4.7[\text{H}]) (2.511[\text{A}])^2 = 14.8 [\text{J}]$$

b.

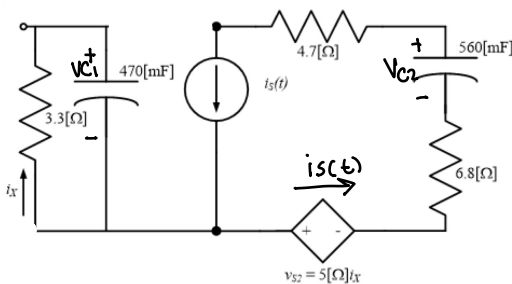
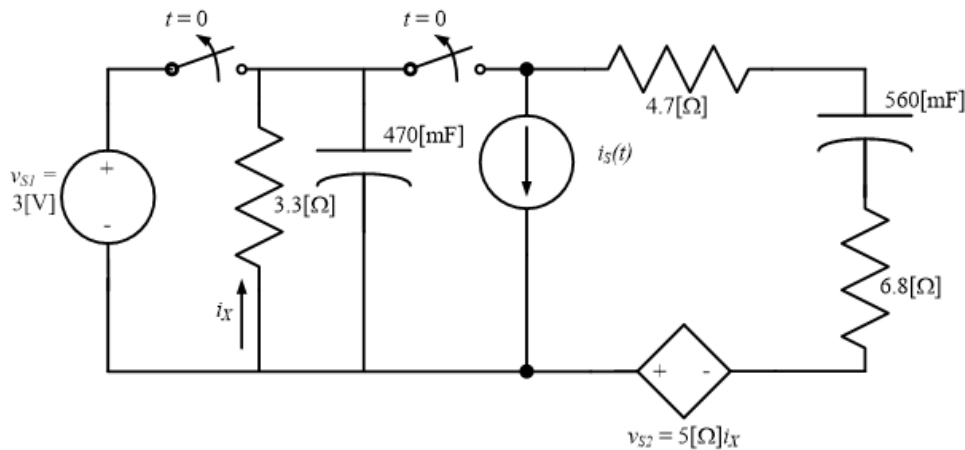
$$P_{\text{del. by } v_s}(4[\text{s}]) = v_s(4[\text{s}]) i_F(4[\text{s}]) = (-7)(4) (-0.976)(4)^2 + 2.1) = 378 [\text{W}]$$

ECE 2202 – Quiz #2

October 10, 2017

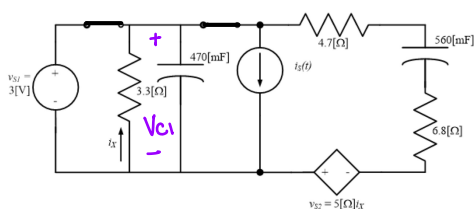
The two switches shown in the circuit below both opened at the same time, at $t = 0$. Find the power delivered by the v_{S2} voltage source at $t = 2[s]$.

$$i_S(t) = 6 \left[\frac{\text{A}}{\text{s}^2} \right] t^2 - 6 [\text{A}].$$



$$i_S(2[s]) = 6(2)^2 - 6 = 18 [\text{A}]$$

$t > 0$



$$v_{C1} = 3[\text{V}] ; v_{C1}(0) = 3[\text{V}]$$

$$\tau = R \cdot C = 3.3[\Omega] \cdot 470[\text{mF}] = 1.551 [\text{s}]$$

formula : $v_C(t) = v_C(0) e^{-t/\tau}$

$t \geq 0$

$$v_C(0) = 3[\text{V}] e^{-t/1.551[\text{s}]}$$

$$i_X(2[s]) = \frac{3e^{-2/1.551[\text{s}]}}{3.3[\Omega]} = -250 [\text{mA}] \quad \checkmark \text{ direction}$$

$$P_{\text{del by } v_{S2}} = (-5[\Omega] i_X) \cdot (i_S) \quad @ 2[s]$$

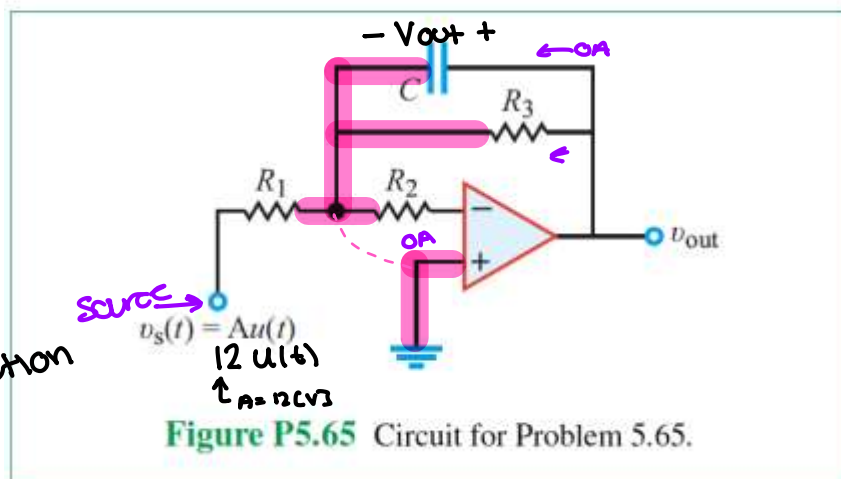
$$P_{\text{del by } v_{S2}} = (-5)(-250\text{mA})(18[\text{A}]) = 22.5 [\text{W}]$$

Textbook #5.65 (Solution in textbook for part b)

5.65 In the circuit of **Fig. P5.65**:

(a) Derive an expression for $v_{out}(t)$ for $t \geq 0$ in terms of R_1 , R_2 , R_3 , C , and A .

* (b) Evaluate the expression for $R_1 = 1 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $C = 0.25 \text{ mF}$, and $A = 12 \text{ V}$.



$$x(t) = \frac{x(0)e^{-at}}{1} + \frac{b}{a}(1 - e^{-at})$$

$$i = C \cdot \frac{dv}{dt}$$

the same portion

$$V_{out}(0^-) = 0 \text{ V}; \quad V_{out}(0) = 0 \text{ V}$$

$$\text{KCL: } \frac{12u(t)[\text{V}]}{R_1} + \frac{V_{out}}{R_3} + C \cdot \frac{dV_{out}}{dt} = 0$$

multiply by $\frac{1}{C}$ to begin to isolate for x' , match formula

$$\frac{12u(t)[\text{V}]}{C \cdot R_1} + \frac{V_{out}}{C \cdot R_3} + \frac{dV_{out}}{dt} = 0$$

$$\text{determine } a \text{ and } b$$

$$a = \frac{1}{C R_3}$$

$$b = -\frac{12u(t)[\text{V}]}{C R_1}$$

Plug and Chug

$$V_{out} = 0 + \frac{-\frac{12u(t)[\text{V}]}{C R_1}}{\frac{1}{C R_3}} (1 - e^{-\frac{1}{C R_3} \cdot t})$$

simplify

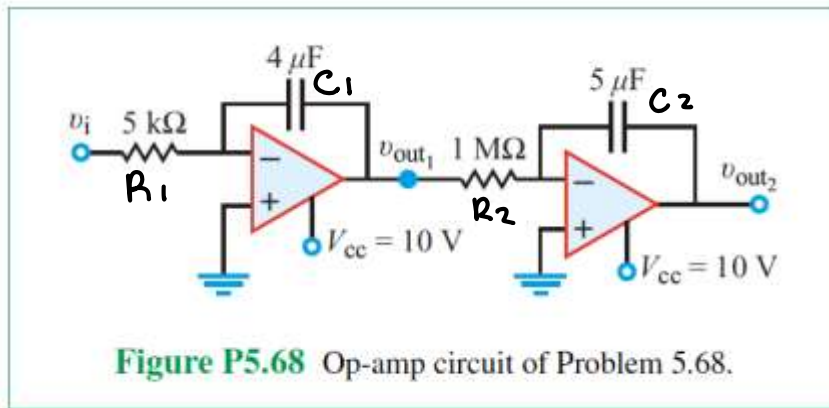
$$V_{out} = \frac{-12u(t)[\text{V}]}{C R_1} \cdot C R_3 (1 - e^{-\frac{1}{C R_3} \cdot t})$$

for $t \geq 0$

Textbook #5.68 (Solution in textbook for part a)

5.68 The two-stage op-amp circuit in **Fig. P5.68** is driven by an input step voltage given by $v_i(t) = 10u(t)$ mV. If $V_{cc} = 10$ V for both op amps and the two capacitors had no charge prior to $t = 0$, determine and plot:

- (a)** $v_{out1}(t)$ for $t \geq 0$;
- (b)** $v_{out2}(t)$ for $t \geq 0$.



a. $v_{out1}(t)$ for $t \geq 0$;

$$\begin{aligned}
 v_i &= 10u(t) \text{ , } V_{cc} = 10 \text{ V} \\
 v_{out1}(t) &= -\frac{1}{R_1 \cdot C_1} \int_0^t v_i dt \\
 &= -\frac{1}{(5 \times 10^3)(4 \times 10^{-6})} \int_0^t 10 dt \\
 &= -50 \int_0^t 10 \text{ [mV]} dv = -0.5 \text{ [V]} \text{ ; for } t > 0
 \end{aligned}$$

b. $v_{out2} = -\frac{1}{R_2 C_2} \int_0^t v_{out1}(t) dt$

$$\begin{aligned}
 R_2 C_2 &= (1 \times 10^6) \cdot (5 \times 10^{-6}) = 5 \\
 &= -\frac{1}{5} \int_0^t (-0.5t) dt = 0.1 \frac{t^2}{2} = 0.05 t^2 \text{ [V]} \text{ } t \geq 0
 \end{aligned}$$